

Laboratory EEG Preprocessing Pipeline

19 people · 32 channels · 500 Hz recording

1 Ingestion

EEG recording

Cz online · Fpz ground

Experiment log

JSONL events · LSL names

Identity map

subject ↔ experiment ↔ XDF

2 Bronze

Raw XDF + logs

immutable inventory

Link and audit

clocks · marker health

Cohort

18 lab · Subject 4 out

3 Silver

Event preprocessing

Marker audit

duplicates · missing

Align event clocks

JSONL ↔ XDF LSL

Canonical markers

observed or reconstructed

EEG signal QC and preprocessing

XDF → MNE Raw

$\mu\text{V} \rightarrow \text{V} \cdot 10^{-20}$ · annotations

Channel QC

condition-blind · QC U review

Notch 50 Hz

line noise

0.5-40 Hz

band-pass

Average ref

exclude bads

Spline interp

spherical

ICA

99% variance · ≤ 3 ocular

4 Gold

Epoch slicing

K_i complete 4 s, each $\mathbb{R}^{32 \times 2000}$

Peak-to-peak reject

any channel above 1050 μV

Welch PSD

2 s windows, 50% overlap

16 features $\in \mathbb{R}^{16}$
Fz θ , posterior α , FAA, Pope (global, FC), Kislov
 δ θ α β γ band dB and relative

Path A — condition windows

5 conditions

$Y_A \in \mathbb{R}^{18 \times 5 \times 16}$

median, IQR, baseline delta

Path B — ad windows

4 ads + 2 no-ad

$Y_B \in \mathbb{R}^{18 \times 6 \times 16}$

post – pre

Shape

T_i : samples for person i . K_i : complete 4 s epochs for person i .

X raw

$\mathbb{R}^{18 \times 32 \times T_i}$

$\tilde{X}$ cleaned

$\mathbb{R}^{18 \times 32 \times T_i}$

E epochs

$\mathbb{R}^{18 \times K_i \times 32 \times 2000}$

$Y_A \in \mathbb{R}^{18 \times 5 \times 16}$

$Y_B \in \mathbb{R}^{18 \times 6 \times 16}$